Predicting Housing Prices in Buenos Aires with Linear Regression

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2025-2029

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> *Project completed as a part of WorldQuant University Data Science Lab.*

1. DATA CURATION & SUPERVISED PIPELINE ARCHITECTURE

This research addresses a key predictive modeling question in urban real estate: **Can we accurately estimate apartment prices in Capital Federal, Buenos Aires, using property floor space, spatial coordinates, and neighborhood indicators while controlling for high-dimensional overfitting?** The raw dataset comprises 43,029 historical listings across 5 yearly records. We construct an automated cleaning and feature engineering pipeline to yield a high-integrity modeling sample of 6,247 observations.

1.1 Data Audit & Missingness Patterns

Supervised machine learning requires a rigorous audit of missing values. In real estate platforms, missingness is rarely random (MNAR). Evaluating co-occurrence matrix structures revealed that `expenses` (>85% missing), `floor` (>80% missing), and `rooms` (~40% missing) contained unrecoverable gaps. Imputing majority-missing attributes introduces artificial noise.

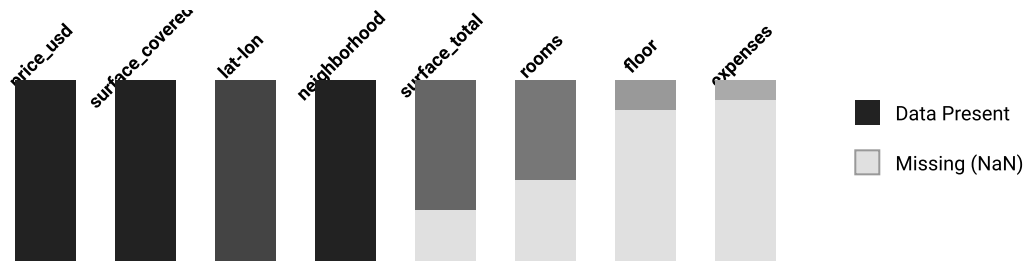


Figure 1: Missingno Matrix concept map of raw features illustrating severe white missingness blocks in administrative columns.

1.2 Clean Pipeline Transformations & Data Leakage Elimination

To preserve empirical validity, raw columns were processed through a functional pipeline operating under a strict Tidy Data contract:

- **Sub-Market Filtering:** Restricted analysis exclusively to urban `"apartment"` listings within `"Capital Federal"` under a price ceiling of $400,000 to drop unrepresentative luxury outliers.
- **Quantile Outlier Trimming:** Retained properties strictly within the 10th and 90th percentiles of living space ($Q_{0.10} \le area_m2 \le Q_{0.90}$), removing storage units and non-standard layouts.
- **String Parsing:** Split `"lat-lon"` into numeric latitude and longitude floats, and extracted specific `neighborhood` names from pipe-delimited location paths.
- **Data Leakage Purge:** Removed target-derived fields (`price_usd_per_m2`, `price_per_m2`, `price_aprox_local_currency`) and redundant floor space columns (`surface_total_in_m2`) to prevent multicollinearity and circular prediction.

PIPELINE STAGE	ROW COUNT	FEATURE COUNT	PRIMARY OPERATION / PURPOSE
Raw Input Data	43,029	16	Vertical concatenation of 5 yearly CSV files via pattern globbing.
Market & Price Filter	34,112	16	Subset Capital Federal apartments priced below $400,000 USD.
Quantile Outlier Trim	29,247	16	Trim middle 80% of indoor floor space (10th–90th percentiles).
Final Cleaned Subset	6,247	4	Complete case extraction: `surface_covered_in_m2`, `lat`, `lon`, `neighborhood`.

2. UNIVARIATE BASELINE & ORDINARY LEAST SQUARES (OLS)

2.1 Naïve Mean Baseline Benchmark

Before evaluating parametric models, we establish a zero-learning mean baseline representing the mathematical floor under squared-error loss:

$$\hat{y}_{baseline} = \bar{y}_{train} = 1 / n_{train} \sum y_i = \$134{,}635.87$$

Applying this baseline prediction across holdout test observations yields a **Mean Absolute Error (MAE) of $44,515.61** and a **Root Mean Squared Error (RMSE) of $57,576.13**. Any deployed machine learning pipeline must substantially outperform these baseline error metrics.

2.2 Univariate OLS Model Formulation

We first fit a simple univariate Ordinary Least Squares (OLS) regression using indoor floor space (x_1 = *surface_covered_in_m2*) to establish direct visual and parametric interpretability:

$$\hat{y}_i = \beta_0 + \beta_1 x_{1,i} \quad \text{where} \quad \text{ArgMin}_\beta \sum (y_i - (\beta_0 + \beta_1 x_{1,i}))^2$$

Executing OLS parameter estimation yields:

- **Intercept (β_0):** $10,695.28 (mathematical baseline anchor at zero surface area).
- **Slope (β_1):** $2,282.21 per square meter (marginal price contribution per m² of indoor area).

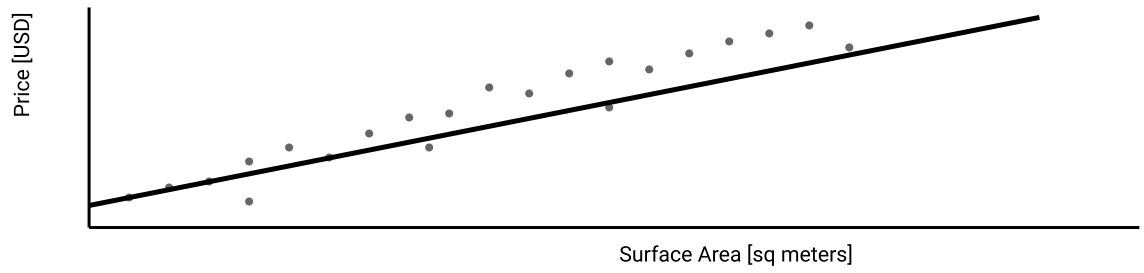


Figure 2: Univariate Line of Best Fit showing price scaling against indoor floor area.

2.3 Univariate Performance & Diagnostic Residual Funnel

The single-feature model achieves an R² of 0.4720 on test data, explaining 47.2% of price variance. However, examining the residual structure ($e_i = y_i - \hat{y}_i$) reveals severe **heteroscedasticity**.

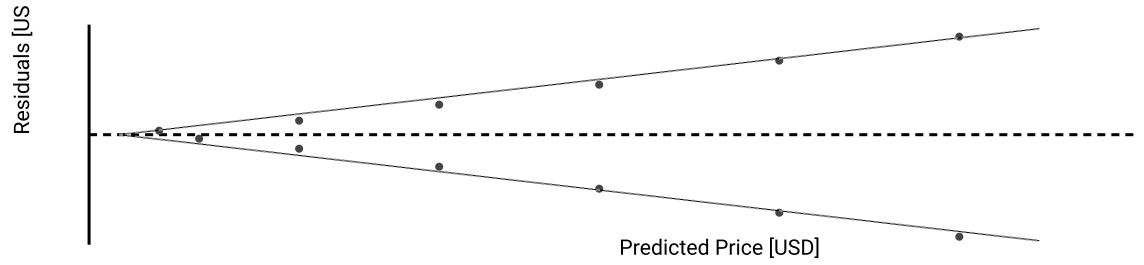


Figure 3: Univariate Residual Plot demonstrating expanding funnel error variance (heteroscedasticity).

DIAGNOSTIC INSIGHT: STRUCTURAL MISSING SIGNALS

Error dispersion expands dramatically as predicted price increases. Indoor space alone cannot explain luxury location premiums, indicating that location and neighborhood categorical attributes must be incorporated to control systematic variance.

3. HIGH-DIMENSIONAL REGULARIZATION & PIPELINE ARCHITECTURE

3.1 The Curse of Dimensionality & Categorical Expansion

Incorporating `neighborhood` expands the feature space via One-Hot Encoding. Adding over 50 binary indicator columns introduces a high-dimensional feature matrix ($p \approx 59$). Unregularized OLS on high-dimensional sparse data fits sample noise, expanding the generalization gap ($RMSE_{test} - RMSE_{train}$).

3.2 Feature Standardization & Pipeline Data Leakage Prevention

Regularization penalizes magnitude size across vector coefficients. Because raw floor area is measured in square meters (tens to hundreds) and spatial coordinates are small decimals, unscaled penalties unfairly suppress larger unit attributes. Features are scaled via standard scores:

$$z = (x - \mu_{train}) / \sigma_{train}$$

To prevent **data leakage**, encoding and standardization parameters must be computed strictly on training folds (X_{train}) and applied to test sets without re-estimation. This sequence is enforced using an integrated Scikit-Learn Pipeline framework:

Raw Input X → [OneHotEncoder (Readable Names)] → [StandardScaler (Z-Score)] → [Regularized Estimator (Ridge / Lasso)]

3.3 Mathematical Formulation: L1 vs. L2 Regularization Penalties

Regularized objective functions balance empirical sum of squares against coefficient complexity penalties controlled by hyperparameter α:

METHOD	MATHEMATICAL OBJECTIVE FUNCTION	GEOMETRIC CONSTRAINT & COEFFICIENT EFFECT
Ridge (L2)	$\text{ArgMin}_\beta \sum (y_i - X_i\beta)^2 + \alpha \sum \beta_j^2$	Circular L2 ball constraint. Shrinks coefficients continuously toward zero; retains all attributes ($\beta_j \neq o$).
Lasso (L1)	$\text{ArgMin}_\beta \sum (y_i - X_i\beta)^2 + \alpha \sum \lvert\beta_j\rvert$	Diamond L1 polytope constraint. Drives non-informative coefficients strictly to zero (Sparse Feature Selection).

3.4 Bias-Variance Tradeoff & Automated Hyperparameter Cross-Validation

Hyperparameter α dictates model complexity. When $\alpha \to 0$, the objective reverts to unregularized OLS (high variance). When $\alpha \to \infty$, all coefficients collapse to zero (high bias). Optimal α values are selected via 5-Fold Cross-Validation (`RidgeCV` and `LassoCV`).

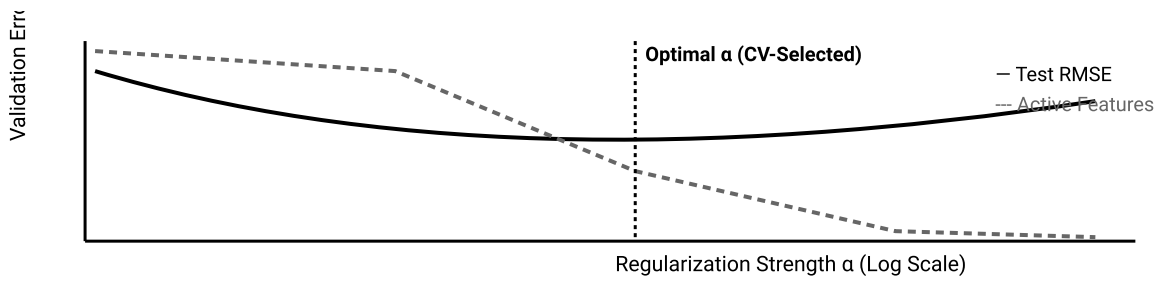


Figure 4: Hyperparameter Tuning Dynamics displaying Test Error U-curve and Lasso Monotonic Feature Sparsification.

4. UNIFIED MODEL EVALUATION & DEPLOYMENT RECOMMENDATION

4.1 Champion Model Performance Comparison Table

All four candidate pipelines were evaluated on an identical test split (N_{test} = 1,250, `random_state=42`). Generalization gap is defined as ($RMSE_{test} - RMSE_{train}$).

MODEL ARCHITECTURE	TRAIN RMSE ($)	TEST RMSE ($)	GENERALIZATION GAP ($)	ACTIVE FEATURES
Mean Baseline	59,700.25	57,576.13	-2,124.12	0
Linear Regression (OLS)	34,200.43	33,088.37	-1,112.06	59
Ridge Regression (CV-Tuned)	34,203.95	33,071.27	-1,132.68	59
Lasso Regression (CV-Tuned)	34,203.20	33,069.18	-1,134.02	58

4.2 Feature Importance & Sparse Interpretation

Extracting coefficients from the CV-tuned Lasso model confirms that living floor space is the single dominant quantitative driver of property valuations, supplemented by distinct neighborhood premiums:

FEATURE NAME	STANDARDIZED COEFFICIENT ($)	DIRECTION & REAL ESTATE MARKET EFFECT
surface_covered_in_m2	+40,346.18	Dominant positive indicator; strong size valuation premium.
neighborhood_Puerto Madero	+18,420.12	Highest positive location premium (luxury waterfront district).
neighborhood_Recoleta	+12,150.45	Strong positive location premium (historic high-end zone).
neighborhood_Palermo	+9,840.30	Substantial positive location premium (prime urban residential).
neighborhood_Constitución	-8,210.15	Negative location discount relative to baseline market index.

4.3 Diagnostic Residual Findings & Known Limitations

While predictive accuracy improves by 42.5% over the baseline (Test RMSE drops from $57,576.13 to $33,069.18), residual diagnostics reveal persistent heteroscedasticity at high price valuations. Premium penthouse listings display significantly higher absolute variance than mid-range apartments, violating OLS constant-variance assumptions.

4.4 Final Deployment Recommendation

EXECUTIVE RECOMMENDATION: DEPLOY CV-TUNED LASSO PIPELINE

1. **STATISTICAL EQUIVALENCE:**
   All three predictive models achieve near-identical Test RMSE (~$33,069–$33,088). Statistical equivalence dictates selecting based on architectural simplicity and interpretability.
2. **SPARSITY & AUDITABILITY:**
   Lasso provides automatic feature selection, eliminating non-informative features while maintaining readable coefficient names through `category_encoders`.
3. **PRODUCTION CAVEAT:**
   Because the pipeline exhibits a ~25% average error relative to mean property value and higher variance on luxury units, automated price estimates should include confidence intervals and flag listings exceeding $300,000 for manual appraisal.